Course: Enterprise Real-Time Data Engineering
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Course Portal: fakhr.me/KafkaSpark
Code & Labs Repository: github.com/fakhruldeen/KafkaSpark

Stack Architecture & Delivery Baseline: Spark 3.5 LTS with Kafka 4.4 KRaft

Classroom Delivery Baseline Decision (Production Certified):
- **Apache Kafka:** 4.4.0 (KRaft mode only; ZK-free)
- **Confluent Platform:** 8.4.0 (Schema Registry, Connect, ksqlDB)
- **Apache Spark / PySpark: 3.5.x LTS** (`pyspark==3.5.9` , `spark-sql-kafka-0-10_2.12:3.5.1` , Java 17 LTS)
- **Status: 100% Certified & Offline Frozen** in `USB_DEPLOYMENT_BUNDLE` . All automated quality gates, validator scripts (`verify_s08.sh` to `verify_s12.sh`), PySpark batch, and Kafka Structured Streaming pipelines are validated against Spark 3.5.x LTS.

Forward Enterprise Migration Roadmap (Post-Course Transition to Spark 4.x):
- Apache Spark 4.x introduces default ANSI SQL mode (`spark.sql.ansi.enabled=true`), Scala 2.13 only, and Python 3.11+.
- The migration notes below detail the verified configuration deltas for enterprise teams planning their forward migration to Spark 4.x in production after course completion.

1. Pins changed in v8

File	v7	v8
`.env` KAFKA_IMAGE	apache/kafka:3.9.0	apache/kafka:4.4.0
`.env` SCHEMA_REGISTRY_IMAGE	cp-schema-registry:7.9.0	cp-schema-registry:8.4.0
`.env` CONNECT_IMAGE	cp-kafka-connect:7.9.0	cp-kafka-connect:8.4.0
`.env` KSQLDB_IMAGE	cp-ksqldb-server:7.9.0	cp-ksqldb-server:8.4.0
`requirements.txt`	pyspark==3.5.1	pyspark==4.3.0
`python/s10_stream.py`, `python/s11_pipeline.py`	spark-sql-kafka-0-10_2.12:3.5.1	spark-sql-kafka-0-10_2.13:4.3.0
All Spark session builders (S08, S09, S10, S11)	no ANSI setting	`spark.sql.ansi.enabled=false` set explicitly

Image tags must be confirmed to exist before the first run (`docker pull` each one). If a patch tag is missing, use the newest published patch in the same line and record it in `.env` .

2. Why the ANSI setting is explicit

Spark 4.0 turns ANSI SQL mode on by default. Results were checked on PySpark 4.3.0 with the S08 batch job:

Run	Read	Clean	Quarantined	Duplicates removed
S08 on Spark 4.3, ANSI off (explicit)	4,033	3,977	23	33
S08 on Spark 4.3, ANSI on (default)	4,033	3,977	23	33
Expected (course reference)	4,033	3,977	23	33

The explicit setting is kept so that the lab behaves the same on any Spark 4 build and matches the Spark 3 behaviour the validated counts were built on. The team may remove it after the S10 and S11 streaming runs pass with ANSI on.

3. Verification so far (sandbox, no Kafka broker)

Lab	Check	Result
S08 batch	Row counts against the course reference	Pass: 4,033 / 3,977 / 23 / 33
S09 batch	Broadcast join plan and revenue table	Pass: BroadcastHashJoin present; 1,985 payments; four regions with the same revenue as v7
S09 batch	Join row count guard	Pass: 3,977 joined rows
All Python files	Syntax check (`py_compile`)	Pass
S10, S11	Structured Streaming with Kafka	Not run: no broker in the sandbox
S01 to S07, S12	Kafka, Connect, Schema Registry, ksqlDB	Not run

Note: the Docker daemon is not available in the build sandbox, so no container stack could be started.

4. Breaking-change check against the official upgrade notes

Source	Change	Effect on the labs	Action
Kafka 4.0 upgrade notes	ZooKeeper removed; KRaft only	Labs already run KRaft	None
Kafka 4.0	Brokers, Connect and tools require Java 17	Lab 00 installs JDK 17	None. Keep JDK 17
Kafka 4.0	`log.message.format.version` and `message.format.version` removed	Not used in the compose file	Check no lab script sets them
Kafka 4.0	`kafka-console-consumer --whitelist` replaced by `--include`; `--bootstrap-server` accepts comma-separated values only	Labs use `--bootstrap-server` and `--timeout-ms`	Run each lab command once
Kafka 4.0	`linger.ms` default changed from 0 to 5	Latency numbers in S03 may move; counts do not	Re-measure any timing in the lab text
Kafka 4.0	`enable.idempotence` no longer falls back when in-flight requests exceed 5	Only matters if a lab sets more than 5 in-flight requests	Check S03 producer config
Kafka 4.0 (checked on the consumer rebalance protocol page)	The server supports the KIP-848 protocol, but clients opt in with `group.protocol=consumer`. The Python labs use the default (classic) protocol and do not set it	No change to S04 or S11 behaviour unless `group.protocol` is set	Confirm that no lab sets `group.protocol`. Session and heartbeat settings for the new protocol are server-side only
Confluent 8.4	`cp-kafka-connect-base` and `cp-server-connect-base` images removed	v8 uses the non-base images	None
Confluent 8.4	`broker.id` deprecated in favour of `node.id`	Compose uses `KAFKA_NODE_ID`	None
Confluent 8.4	`_confluent-license` topic deprecated; `confluent.license` key	Not set in the lab compose file	Confirm the images start without a licence key; set `confluent.license.topic` if a warning blocks start
Confluent 8.4	`kafka-storage format` fails on a formatted directory unless `--ignore-formatted`	Only if a volume is reused	Use `make reset` before the first v8 run

Source	Change	Effect on the labs	Action
Confluent 8.4	Schema Registry `LOGICAL` compatibility conflicts with `FORWARD` and `FORWARD_TRANSITIVE`	S05 uses BACKWARD	Confirm S05 uses BACKWARD
Confluent 8.4	jmx_prometheus_javaagent 1.6.0; metrics path `/metrics`	Only extra module X7	Update scrape path when X7 is run
Spark 4.0	ANSI mode on by default	Handled by explicit setting (section 2)	None
Spark 4.0	`spark.sql.maxSinglePartitionBytes` default 128m	Affects single-file writes only	Check S08 output file counts
Spark 4.0	ORC default codec zstd	Labs write Parquet	None
Spark 4.0	Scala 2.13 only	Kafka connector coordinate changed to `_2.13`	Done in v8
Spark 4.0	Python 3.11 or later	Lab 00 must install Python 3.11+	Update Lab 00
Spark 4.0	Timestamp parse policy `CORRECTED` by default	Affects `to_timestamp` on malformed input	Check quarantine count after S11 run

5. Checked and open items

Checked against the official pages on 2026-11-03:

- Spark Kafka connector coordinate for 4.3.0 is `org.apache.spark:spark-sql-kafka-0-10_2.13:4.3.0`. Matches v8.
- Spark Kafka options used by the labs (`startingOffsets`, `failOnDataLoss`, `maxOffsetsPerTrigger`) are valid in 4.3.0. None of the options listed as blocked (`group.id`, `auto.offset.reset`, `enable.auto.commit`, deserializers) is set in a Spark lab. The Python consumer labs (S04, S05) set them on the confluent-kafka client, which is not affected.
- Kafka 4.0 consumer rebalance protocol: opt-in per client (correction to the earlier note in this file).

Still open, must be read before the run:

- Kafka 4.x KRaft settings for `node.id` and controller quorum in the upgrade guide (the v8 compose file uses the same keys as v7).
- Kafka Streams upgrade guide for 4.x (affects the Java take-home notes only).
- Confluent 8.4 Schema Registry and ksqlDB release notes beyond the items above.

- Kafka client version bundled with the Spark 4.3 connector (Spark page does not state it); check the POM after the Maven download is possible.

Network note: the build sandbox cannot reach Docker Hub or Maven Central (policy denial), so the images and Maven artifacts could not be pulled here. The image tags are therefore unconfirmed.

6. Run order for the validation

1. `make reset` on a clean host; pull the four images; confirm tags exist.
2. Start the stack and run `make check`.
3. S01 to S04 labs and their PASS checks; CP1.
4. S05 to S07 labs; CP2.
5. S08 to S10 labs; CP3.
6. S11 lab and acceptance tests T1 to T5; CP4.
7. Record every count and timing in the trainer log, and compare with the v7 reference values.
8. Only after steps 1 to 7 pass, mark v8 as the delivery stack in Lab 00 and in the Phase 1 environment list.

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